

Prerequisites

- ⚠️ **Important**
- Use Kiro CLI for this section
 - Complete [Setting Up Kiro](#)

Sample 1: Schedule EC2 with Kiro CLI

1. In your VS Code Server environment, create a new folder to separate context from other files inside VS Code Server:

```
1 cd /Workshop
2 mkdir ec2-schedule
3 cd ec2-schedule
```



2. Use the following prompt.

```
1 Create an automation using AWS Systems Manager
2 To schedule EC2 instances with names containing 'sampleapp-load-generator' in us-east-1
3 The EC2 instances should be running on weekdays, 9-to-5 Bangkok time, and stopped during other time slots
```



EC2 Scheduler for sampleapp-load-generator Instances

This project sets up an automated scheduling system for EC2 instances with names containing 'sampleapp-load-generator' in the us-east-1 region. The instances will automatically start on weekdays at 9 AM Bangkok time and stop at 5 PM Bangkok time.

Components

- **CloudFormation Template:** Defines all the necessary AWS resources
- **Lambda Function:** Handles the starting and stopping of EC2 instances
- **EventBridge Rules:** Schedules the Lambda function execution
- **IAM Role:** Provides necessary permissions for the Lambda function

Time Conversion

- Bangkok time (ICT) is UTC+7
- 9 AM BKK = 2 AM UTC
- 5 PM BKK = 10 AM UTC

Deployment

Sample 2: CloudTrail Analysis

1. In your VS Code Server environment, create a new folder to separate context from other files inside VS Code Server:

```
1 cd /Workshop
2 mkdir cloudtrail-analysis
3 cd cloudtrail-analysis
```



2. Within the VS Code Server environment, start Kiro CLI by running:

```
1 kiro-cli chat
```



3. Use the following prompt. Don't forget to input your email address  :

```
1 Analyze 24-hour periods of CloudTrail data and compare them
2 These are the minimum dimensions I'm looking for:
3 1. Comprehensive pattern analysis
4 2. Anomaly detection
5 3. Actionable insights on what needs attention
6 All formatted in a clean, digestible report.
7 Please send it via SNS topic named cloudtrail-report every morning at 8 AM Thailand time. Add $YOUR_EMAIL as a subscriber
```



CloudTrail Analysis Solution

This solution analyzes CloudTrail data over 24-hour periods, identifies patterns, detects anomalies, and generates comprehensive reports that are delivered via SNS.

Features

- **Comprehensive Pattern Analysis:** Analyzes CloudTrail events to identify usage patterns across services, regions, users, and event types
- **Anomaly Detection:** Identifies unusual activities such as sensitive operations, unusual source IPs, and significant changes in event volume
- **Actionable Insights:** Provides recommendations based on the analysis results
- **Automated Reporting:** Sends daily reports via SNS at 8 AM Thailand time

Setup Instructions

1. Make sure you have AWS CLI configured with appropriate permissions:

- CloudTrail: `LookupEvents`
- SNS: `CreateTopic`, `Subscribe`, `Publish`

2. Run the setup script:

```
chmod +x setup_cloudtrail_analysis.sh
./setup_cloudtrail_analysis.sh
```

3. Confirm the SNS subscription by clicking the link in the confirmation email sent to th-team@amazon.com

Report Contents

Each daily report includes: